Brett Adams

Software engineering student building backend services, data pipelines and developer tools
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Education

Honours Computer Science with Software Engineering Specialization Co-op **Sep 2022 - 2027 (expected)**
University of Windsor *Windsor, ON*

Skills

Programming Languages	Python | TypeScript | JavaScript | SQL | Java | C++ | C | HTML/CSS | VBA
Software & Tools	Node.js | Flask | React | Git | GitHub Actions | GitLab CI/CD | Docker
Data & Cloud	BigQuery | Apache Airflow | Cloud Composer | dbt | DuckDB | Azure Databricks | Power BI

Experience

Data Analyst Intern **May 2026 - Present**
Geotab *Oakville, ON*

- Maintain and debug production Apache Airflow DAGs on Cloud Composer that run the SalesOps data pipelines.
- Write and deploy production BigQuery SQL for Revenue Operations tables, including the device status, database dimension and accounts master datasets.
- Ship changes through GitLab CI/CD, from branch development through staging validation to production.
- Carry out live table migrations against fixed infrastructure deadlines, coordinating with dataset owners.

Data Management Associate, Co-op **Jan 2025 - Aug 2025**
Ontario Ministry of Transportation *St. Catharines, ON*

- Built Power BI dashboards on Azure Databricks and SQL that replaced manual reporting across business units.
- Automated legacy workflows with SQL queries and VBA scripts, cutting processing time from hours to minutes.
- Led the rollout of the internal Data Broker Tool, onboarding teams and standardizing the data vocabulary.

Digital Solutions, Co-op **Apr 2024 - Sep 2024**
CIMCO Refrigeration *Burlington, ON*

- Built Flask web applications that deliver real-time data and alarms to NHL arena operations teams, including a thermal imagery viewer used by arena technicians.
- Wrote a Python pipeline that aggregates data from two APIs into structured JSON for the arena interface.
- Automated controller data collection for CAG engineers with a Python tool, replacing manual data entry.
- Designed and ran hands-on Python training sessions for over 20 employees.

Digital Solutions **Apr 2023 - Sep 2023**
CIMCO Refrigeration *Burlington, ON*

- Built Power Platform tools that several departments adopted to automate repetitive manual workflows.
- Automated SharePoint document routing and approvals, cutting turnaround time between departments.

Notable Projects

border-waits-pipeline **Oct 2026 - Present**
Python, dbt and DuckDB pipeline over Canadian and U.S. border wait-time feeds *Independent*

- Collects the wait estimates CBSA and U.S. CBP publish, models them with dbt on DuckDB, and publishes Parquet that a live dashboard queries with SQL in the browser.
- Reconciled two feeds that disagree about time: CBP's "At Noon EDT" means the hour about to start, and the two sides of one crossing sit in different time zones.
- A failed run redeploys the last good export with a status file naming the step that broke, so a failed refresh keeps the previous data and reports the failure. Tests run on recorded fixtures with no network.

mcp-surface **Jul 2026 - Present**
TypeScript CLI and GitHub Action for testing Model Context Protocol servers *Independent*

- Published on npm. Connects to an MCP server over stdio, Streamable HTTP or SSE the way a real client does, and runs 19 checks on the tools, resources and prompts it actually exposes.
- Snapshots the tool surface and fails CI when it changes unexpectedly: a Zod 3 to 4 upgrade passed every unit test in one of my servers, and mcp-surface failed the build because all 23 tool schemas had changed.

MCP Servers for Claude Code **Jul 2026 - Present**
Five Node.js servers exposing 69 tools to AI assistants *Independent*

- Covers Google Workspace (23 tools, on a self-registered OAuth client that requests only non-restricted scopes), Reddit, Yahoo Fantasy, Chess.com and three Supercell game APIs. Tests run with no network or credentials.
- Hardened against untrusted model output: a subject line that smuggles in a Bcc header is refused, and spreadsheet values are written as plain text unless formulas are requested. Each case has a test that builds the attack.